

PRACTICAL 7

2520090088-B.DHARSHAN

7. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
root@PC: ~/memlayout
mv: cannot stat '/var/run/motd.new': No such file or directory
ERROR: could not install new MOTD

This message is shown once a day. To disable it please create the
/root/.hushlogin file.
root@PC:~# mkdir ~/memlayout && cd ~/memlayout
root@PC:~/memlayout#
root@PC:~/memlayout# nano memlayout.c
root@PC:~/memlayout# gcc -o memlayout memlayout.c
root@PC:~/memlayout# ./memlayout
Address of main (code segment): 0x555e198f1212
Address of code (function): 0x555e198f11e9
Address of global_var (data): 0x555e198f4010
Address of global_uninit (bss): 0x555e198f401c
Address of static_var (data): 0x555e198f4014
Address of static_uninit (bss): 0x555e198f4020
Address of heap_var (heap): 0x555e494172a0
Address of stack_var (stack): 0x7ffeeca7bdcc

PID of this process: 665
Sleeping 30s so you can inspect /proc/665/maps ...
sudo apt install procps -y
root@PC:~/memlayout# sudo apt install procps -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64
  libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libproc2-0
The following packages will be upgraded:
  libproc2-0 procps
2 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
Need to get 766 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libproc2-0 amd64 2:4
.0.4-4ubuntu3.3 [58.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 procps amd64 2:4.0.4-4
```



```

root@PC: ~/memlayout
sudo apt install procps -y
root@PC:~/memlayout# sudo apt install procps -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64
  libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libproc2-0
The following packages will be upgraded:
  libproc2-0 procps
2 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
Need to get 766 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libproc2-0 amd64 2:4.0.4-4ubuntu3.3 [58.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 procps amd64 2:4.0.4-4ubuntu3.3 [707 kB]
Fetched 766 kB in 10s (75.3 kB/s)
(Reading database ... 48003 files and directories currently installed.)
Preparing to unpack .../libproc2-0_2%3a4.0.4-4ubuntu3.3_amd64.deb ...
Unpacking libproc2-0:amd64 (2:4.0.4-4ubuntu3.3) over (2:4.0.4-4ubuntu3.2) ...
Preparing to unpack .../procps_2%3a4.0.4-4ubuntu3.3_amd64.deb ...
Unpacking procps (2:4.0.4-4ubuntu3.3) over (2:4.0.4-4ubuntu3.2) ...
Setting up libproc2-0:amd64 (2:4.0.4-4ubuntu3.3) ...
Setting up procps (2:4.0.4-4ubuntu3.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
root@PC:~/memlayout# pmap -x <PID>
-bash: syntax error near unexpected token `newline'
root@PC:~/memlayout# size memlayout
   text    data     bss      dec     hex filename
   2651     648      16    3315     cf3 memlayout
root@PC:~/memlayout# objdump -h memlayout

memlayout:      file format elf64-x86-64

Sections:
```

		CONTENTS, ALLOC, LOAD, READONLY, DATA				
6	.dynstr	000000c3	00000000000004f8	00000000000004f8	000004f8	2**0
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
7	.gnu.version	00000018	00000000000005bc	00000000000005bc	000005bc	2**1
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
8	.gnu.version_r	00000040	00000000000005d8	00000000000005d8	000005d8	2**3
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
9	.rela.dyn	000000c0	0000000000000618	0000000000000618	00000618	2**3
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
10	.rela.plt	00000090	00000000000006d8	00000000000006d8	000006d8	2**3
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
11	.init	0000001b	0000000000001000	0000000000001000	00001000	2**2
		CONTENTS, ALLOC, LOAD, READONLY, CODE				
12	.plt	00000070	0000000000001020	0000000000001020	00001020	2**4
		CONTENTS, ALLOC, LOAD, READONLY, CODE				
13	.plt.got	00000010	0000000000001090	0000000000001090	00001090	2**4
		CONTENTS, ALLOC, LOAD, READONLY, CODE				
14	.plt.sec	00000060	00000000000010a0	00000000000010a0	000010a0	2**4
		CONTENTS, ALLOC, LOAD, READONLY, CODE				
15	.text	00000289	0000000000001100	0000000000001100	00001100	2**4
		CONTENTS, ALLOC, LOAD, READONLY, CODE				
16	.fini	0000000d	000000000000138c	000000000000138c	0000138c	2**2
		CONTENTS, ALLOC, LOAD, READONLY, CODE				
17	.rodata	00000183	0000000000002000	0000000000002000	00002000	2**3
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
18	.eh_frame_hdr	0000003c	0000000000002184	0000000000002184	00002184	2**2
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
19	.eh_frame	000000cc	00000000000021c0	00000000000021c0	000021c0	2**3
		CONTENTS, ALLOC, LOAD, READONLY, DATA				
20	.init_array	00000008	0000000000003d90	0000000000003d90	00002d90	2**3
		CONTENTS, ALLOC, LOAD, DATA				
21	.fini_array	00000008	0000000000003d98	0000000000003d98	00002d98	2**3
		CONTENTS, ALLOC, LOAD, DATA				
22	.dynamic	000001f0	0000000000003da0	0000000000003da0	00002da0	2**3
		CONTENTS, ALLOC, LOAD, DATA				
23	.got	00000070	0000000000003f90	0000000000003f90	00002f90	2**3
		CONTENTS, ALLOC, LOAD, DATA				
24	.data	00000018	0000000000004000	0000000000004000	00003000	2**3
		CONTENTS, ALLOC, LOAD, DATA				
25	.bss	00000010	0000000000004018	0000000000004018	00003018	2**2

```
root@PC: ~/memlayout
ALLOC
26 .comment 0000002d 0000000000000000 0000000000000000 00003018 2**0
CONTENTS, READONLY
root@PC:~/memlayout# readelf -l memlayout

Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1100
There are 13 program headers, starting at offset 64

Program Headers:
Type           Offset             VirtAddr           PhysAddr
               FileSiz            MemSiz             Flags   Align
PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
               0x00000000000002d8 0x00000000000002d8 R       0x8
INTERP         0x0000000000000318 0x0000000000000318 0x0000000000000318
               0x000000000000001c 0x000000000000001c R       0x1
  [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
               0x0000000000000768 0x0000000000000768 R       0x1000
LOAD           0x0000000000000100 0x0000000000000100 0x0000000000000100
               0x0000000000000399 0x0000000000000399 R E     0x1000
LOAD           0x0000000000000200 0x0000000000000200 0x0000000000000200
               0x000000000000028c 0x000000000000028c R       0x1000
LOAD           0x00000000000002d9 0x00000000000003d9 0x00000000000003d9
               0x0000000000000288 0x0000000000000298 RW      0x1000
DYNAMIC         0x00000000000002da 0x00000000000003da 0x00000000000003da
               0x00000000000001f0 0x00000000000001f0 RW      0x8
NOTE           0x0000000000000338 0x0000000000000338 0x0000000000000338
               0x0000000000000030 0x0000000000000030 R       0x8
NOTE           0x0000000000000368 0x0000000000000368 0x0000000000000368
               0x0000000000000044 0x0000000000000044 R       0x4
GNU_PROPERTY   0x0000000000000338 0x0000000000000338 0x0000000000000338
               0x0000000000000030 0x0000000000000030 R       0x8
GNU_EH_FRAME   0x0000000000000218 0x0000000000000218 0x0000000000000218
               0x000000000000003c 0x000000000000003c R       0x4
GNU_STACK      0x0000000000000000 0x0000000000000000 0x0000000000000000
               0x0000000000000000 0x0000000000000000 RW      0x10
GNU_RELRO      0x00000000000002d9 0x00000000000003d9 0x00000000000003d9
               0x0000000000000270 0x0000000000000270 R       0x1
```

root@PC: ~/memlayout

LOAD	0x0000000000001000	0x0000000000001000	0x0000000000001000	
	0x000000000000399	0x000000000000399	R E	0x1000
LOAD	0x0000000000002000	0x0000000000002000	0x0000000000002000	
	0x00000000000028c	0x00000000000028c	R	0x1000
LOAD	0x0000000000002d90	0x0000000000003d90	0x0000000000003d90	
	0x000000000000288	0x000000000000298	RW	0x1000
DYNAMIC	0x0000000000002da0	0x0000000000003da0	0x0000000000003da0	
	0x0000000000001f0	0x0000000000001f0	RW	0x8
NOTE	0x000000000000338	0x000000000000338	0x000000000000338	
	0x000000000000030	0x000000000000030	R	0x8
NOTE	0x000000000000368	0x000000000000368	0x000000000000368	
	0x000000000000044	0x000000000000044	R	0x4
GNU_PROPERTY	0x000000000000338	0x000000000000338	0x000000000000338	
	0x000000000000030	0x000000000000030	R	0x8
GNU_EH_FRAME	0x0000000000002184	0x0000000000002184	0x0000000000002184	
	0x00000000000003c	0x00000000000003c	R	0x4
GNU_STACK	0x0000000000000000	0x0000000000000000	0x0000000000000000	
	0x0000000000000000	0x0000000000000000	RW	0x10
GNU_RELRO	0x0000000000002d90	0x0000000000003d90	0x0000000000003d90	
	0x000000000000270	0x000000000000270	R	0x1

Section to Segment mapping:

Segment Sections...

00	
01	.interp
02	.interp .note.gnu.property .note.gnu.build-id .note.ABI-tag .gnu.hash .dynsym .dynstr .gnu.version .gnu.version_r .rela.dyn .rela.plt
03	.init .plt .plt.got .plt.sec .text .fini
04	.rodata .eh_frame_hdr .eh_frame
05	.init_array .fini_array .dynamic .got .data .bss
06	.dynamic
07	.note.gnu.property
08	.note.gnu.build-id .note.ABI-tag
09	.note.gnu.property
10	.eh_frame_hdr
11	
12	.init_array .fini_array .dynamic .got

root@PC:~/memlayout#